Systems and Inequalities

SAT Math · Foundations to Mastery · Week of 11 October

Name: _____ Due: _____

How to use this sheet

This is four short sittings, not one long one. Four half-hours across the week will do more for you than one long Saturday, and you are far more likely to finish it.

- **Part A is no calculator.** None. It is 15 minutes and it keeps the last two weeks alive.
- **In Part B, name your method before you start.** Every time. Choosing the cheapest route is the skill being graded here, not grinding.
- When you get one wrong, write it in your error log: what you did, and what you should have done.

Sitting	Part	What it drills	Minutes
Sunday	A	Retention: signs and slope	15
Tuesday	B	Solve the system, and name the method	25
Thursday	C	Inequalities, including translation	25
Saturday	D, E	Build the system from words; six mixed, timed	25

Part A — Retention

No calculator. 15 minutes.

SUNDAY

The first five keep last fortnight's sign work alive. The last five keep week two's lines alive. Short answers. If a sign is in doubt, draw a number line in the margin.

1. $8-(-5)=$ _____
2. $(-4)^2-4^2=$ _____
3. Expand: $-2(x-6)=$ _____
4. $f(x)=x^2-3x$. Find $f(-2)$: _____
5. $-(5-9)=$ _____
6. A line passes through $(-2,7)$ and $(4,-5)$. Its slope is _____
7. For $3x-2y=12$, the y-coordinate of the y-intercept is _____
8. For $3x-2y=12$, the slope is _____
9. For $y=4-0.5x$, the slope is _____
10. A repair costs $C(t)=45+18t$ dollars for t hours of labour. In one short phrase, what does the 18 represent?

    _____

Part B — Solve the system, and name the method

Calculator allowed. 25 minutes.

TUESDAY

You have three tools: substitution, elimination, and graphing both lines on Desmos. For each one, write the answer *and* write which tool you chose. Choosing is a decision, and on a timed test the cheapest tool is worth as much as the arithmetic. Look at the system for five seconds before you pick up your pen.

1. $y = 2x - 3$ and $4x + y = 15$. What is the value of x?
   Answer: _____ *(write your answer)*
   Method: _____ *(substitution, elimination, or graphing)*

2. $x = 4 - 3y$ and $2x + 5y = 3$. What is the value of y?
   Answer: _____ *(write your answer)*
   Method: _____ *(substitution, elimination, or graphing)*

3. $4x + 3y = 18$ and $4x - 5y = 2$. What is the value of y?
   Answer: _____ *(write your answer)*
   Method: _____ *(substitution, elimination, or graphing)*

4. $3x + 2y = 7$ and $5x - 2y = 17$. What is the value of y?
   Answer: _____ *(write your answer)*
   Method: _____ *(substitution, elimination, or graphing)*

5. $0.6x + 1.5y = 6.9$ and $2.4x - 0.5y = 8.1$. What is the value of x?
   Answer: _____ *(write your answer)*
   Method: _____ *(substitution, elimination, or graphing)*

6. $7x + 11y = 101$ and $11x + 7y = 97$. What is the value of $x + y$?
   Answer: _____ *(write your answer)*
   Method: _____ *(substitution, elimination, or graphing)*

7. How many solutions does the system $2x - 6y = 9$ and $x - 3y = 7$ have?
   A) Zero B) Exactly one C) Exactly two D) Infinitely many
   Method: _____ *(substitution, elimination, or graphing)*

8. How many solutions does the system $3x + 4y = 10$ and $9x + 12y = 30$ have?
   A) Zero B) Exactly one C) Exactly two D) Infinitely many
   Method: _____ *(substitution, elimination, or graphing)*

Part C — Inequalities

Calculator allowed. 25 minutes.

THURSDAY

Two things are being tested: the flip when you multiply or divide by a negative, and reading an English phrase into a symbol. After every solve, test one value on each side of the boundary. That check costs ten seconds and catches a missing flip every time.

1. Which inequality gives all values of x for which $5 - 3x \geq 17$?
   A) $x \geq -4$ B) $x \leq -4$ C) $x \leq 4$ D) $x \geq 4$

2. What is the greatest integer x for which $4 - 7x \geq -10$?
   Answer: _____ *(write your answer)*

3. Which inequality gives all values of x for which $-2(x - 5) > 14$?
   A) $x < -2$ B) $x > -2$ C) $x < -12$ D) $x > -12$

4. A game gives you 4 points for each task you finish. You already have 14 points and you need a total of at least 50. Which inequality describes the number of tasks t that is enough?
   A) $14 + 4t > 50$ B) $14 + 4t \geq 50$ C) $14 + 4t \leq 50$ D) $4t \geq 50$

5. A food truck charges \$3 for a taco and \$5 for a burrito. On one shift it took in no more than \$150 from these two items. Which combination is possible?
   A) 20 tacos and 20 burritos B) 30 tacos and 12 burritos
   C) 40 tacos and 8 burritos D) 10 tacos and 25 burritos

6. Five less than the product of -4 and a number is more than 23. Which inequality describes all such numbers n?
   A) $n < -7$ B) $n > -7$ C) $n < -4.5$ D) $n > -4.5$

7. On every day of a week the high temperature stayed between 12 and 19 degrees, and it never reached either of those two values. Which inequality describes the high temperature t?
   A) $12 \leq t \leq 19$ B) $12 < t < 19$ C) $12 > t > 19$ D) $t > 12$ or $t < 19$

8. A line is drawn on the xy-plane through $(0, -3)$ with slope 2. The line itself is dashed, and the whole region *above* it is shaded. Which inequality does the shaded region represent?
   A) $y > 2x - 3$ B) $y \geq 2x - 3$ C) $y < 2x - 3$ D) $y \leq 2x - 3$

Part D — Build the system from words

No calculator. 15 minutes.

SATURDAY

Write the two equations only. Do not solve them. That is the whole task. Setting up the pair is the step that costs you marks, so this part isolates it and nothing else. Say in words what each letter stands for, then write the two equations.

1. You buy 14 items in total, some pens at \$2 each and some notebooks at \$5 each, and you spend \$43 altogether. Let p be the number of pens and n the number of notebooks.

2. Two numbers have a sum of 56. One of them is 8 more than twice the other. Let x be the larger number and y the smaller.

3. A school sold 160 tickets to a concert. Adult tickets were \$12 and child tickets were \$7, and the takings were \$1,395. Let a be the number of adult tickets and c the number of child tickets.

4. A rectangular garden has a perimeter of 86 metres. Its length is 7 metres more than twice its width. Let L be the length and W the width, both in metres.

5. Maya has 18 coins, all dimes and quarters, worth \$3.15 in total. Let d be the number of dimes and q the number of quarters.

Part E — Mixed, timed

Calculator allowed. 12 minutes, timed.

SATURDAY

Set a timer for 12 minutes. These span Parts B to D. Answer every one — there is no penalty for a wrong answer, so a blank is a mark thrown away. Then check each answer by substituting it back into both equations.

1. $y = 3x - 8$ and $2x + y = 7$. What is the value of y?
   Answer: _____ *(write your answer)*

2. $6x + 5y = 37$ and $5x + 6y = 29$. What is the value of $x + y$?
   Answer: _____ *(write your answer)*

3. Which inequality gives all values of x for which $9 - 4x \le 29$?
   A) $x \ge -5$ B) $x \le -5$ C) $x \ge 5$ D) $x \le 5$

4. A theatre sold 90 tickets in total. Adult tickets cost \$15 and student tickets cost \$9, and the takings were \$1,062. How many adult tickets were sold?
   Answer: _____ *(write your answer)*

5. A tutor charges a \$40 booking fee plus \$28 for each hour. A family wants the total to be no more than \$180. Which inequality describes the number of hours h they can book?
   A) $40 + 28h < 180$ B) $40 + 28h \le 180$ C) $28h \le 180$ D) $40 + 28h \ge 180$

6. In the system $2x + ky = 10$ and $6x + 9y = 30$, the constant k is chosen so that the system has infinitely many solutions. What is the value of k?
   Answer: _____ *(write your answer)*

Error log

Fill this in as you go.

Date	Q	What I did	What I should have done

Bring this sheet and your error log to the next session.